ROAM: A Retrieval-augmented Optimization Agentic Model for Spontaneous Trip Blueprint Generation

SYE6621 Agentic AI for Innovation

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Background & Motivation

01 · PROBLEM

Why this problem matters

- Travel planning is hard — a manual itinerary costs 3–5 hours of research and trade-offs.
- Post-pandemic, people want spontaneity, but work pressure ("involution") leaves little time.
- So we need planning that is fast, flexible, and easy to change on the fly.

Why pure LLMs fail

1. Spatial cognitive bias

Hallucinated distances, locations, and travel times between POIs.

2. Time-window ignorance

Ignores opening hours and realistic daily schedules.

3. Knowledge lag

Outdated venue information, no live retrieval.

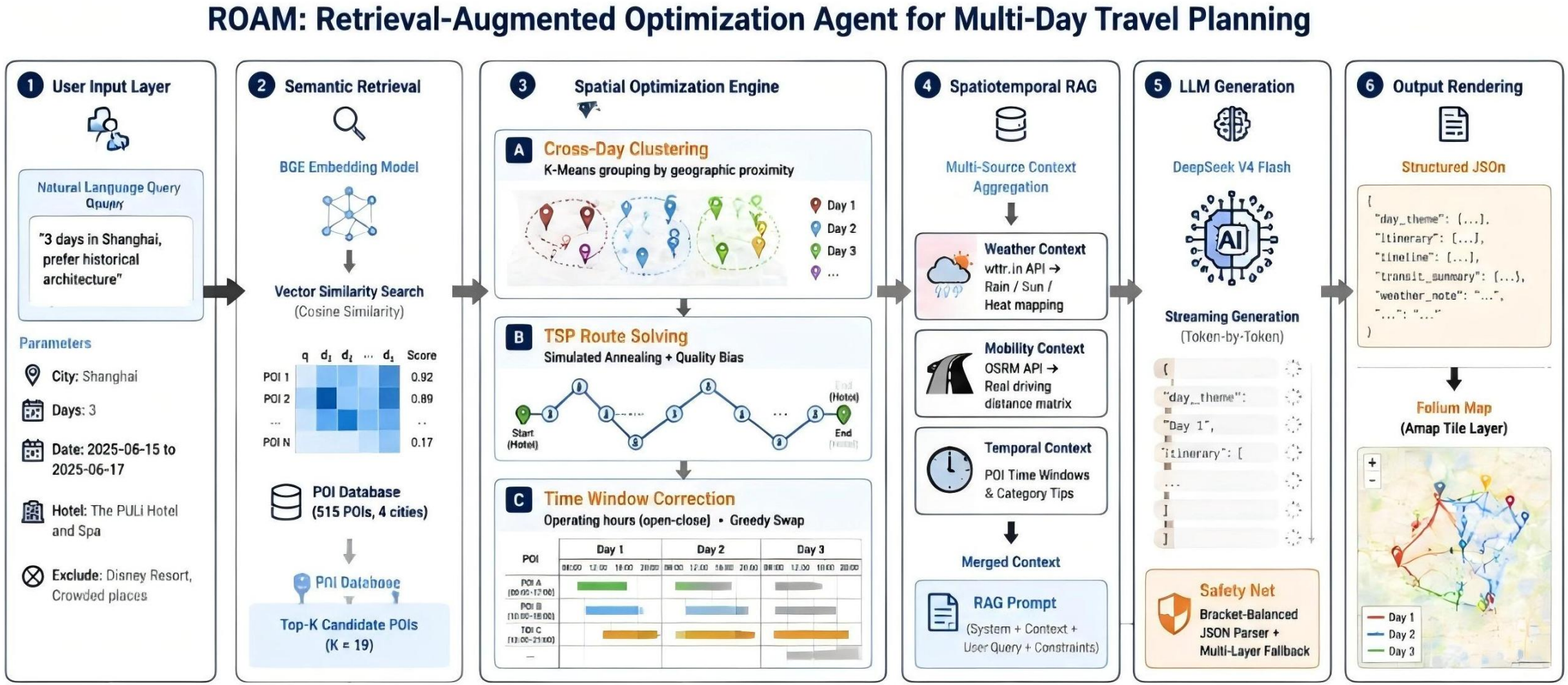
Prior Works & Positioning

02 · PRIOR WORK

Prior Work	Core Idea	Strength	Limitation	Our Relevance
ITINERA (Tang et al., KDD UrbComp '24)	Retrieval + LLM with a POI knowledge graph	Strong semantic personalization	Single-day; weak hard time-windows	Extend to multi-day + replanning
Commercial platforms (Ctrip / Mafengwo)	Heuristics + manual editing UI	Huge POI coverage; usable	No autonomous NL revision; rigid	Add LLM-driven autonomous edits
OR methods (OP / TSPTW)	Exact route optimization	Provably optimal routes	No language understanding; fixed sets	Keep optimizer as a tool in LLM loop

→ **We target the multi-day scope, hard time-window feasibility, and category-aware dwell time left open by prior works.**

Proposed Method



The initial pipeline produces a static multi-day plan.

Replanning re-invokes stages 2–5 on demand whenever the user requests a change.

Proposed Method Replanning Agent

03 · METHOD

The problem

- The initial plan is static — but real trips are not.
- Users want on-the-fly edits in plain language:

  “Swap the park on Day 1 afternoon for a museum.”

- Manual re-planning loses ROAM’s optimizer guarantees.

The solution

- A Replanner Agent driven by LLM function calling (ReAct-style).
- The LLM autonomously chooses which tools to call to satisfy the request.
- Edits run through the same optimizers used at planning time.
- Result: natural-language revision with planner-grade feasibility.

Example

← Return to dialogue

🎯 A Three-Day Trip to Fuzhou: Alleys, Riverside, and Everyday Life

Three days of immersive experience of Fuzhou's ancient charm and modern style: on the first day, stroll through the Three Hills, Two Pagodas, and Three Alleys and Seven Alleys, savoring authentic snacks; The next day, visit the modern architectural complex of Nantai Island to appreciate the new scenery along both banks of the Min River; On the last day, wandering through ancient streets and bustling streets, experiencing the perfect blend of tradition and everyday life.

Day 1: 三山两塔·坊巷寻味 | Day 2: Modern Memories and New Riverside Scenery of Nantai Island | Day 3: Ancient Charm and Modern Style · A Journey Through the Everyday Life of the City

📍 Route

Straits Cultural and Arts Center - > Flower Sea Park - > Taijiang Yacht Marina - > Yantai Mountain - > Fujian Normal University (Cangshan Campus) - > Fuyuan Road - > Gaogai Mountain Park - > Luozhou Ancient Town

Connecting modern cultural landmarks, the century-old consulate's charm, and the ancient town bustle, the half-day experience condenses Fuzhou's cultural depth and warmth of life.

🚗 Transportation

🚗 Transportation advice and cost breakdown

Straits Cultural and Arts Center → Flower Sea Park: 🚗 3103m by car (about 3 minutes, taxi about 19 RMB) Huahai Park → Taijiang Yacht Marina: 🚲 Cycling 1491m (about 5 minutes, or about 18 minutes on foot) Taijiang Yacht Marina → Yantai Mountain: 🚗 Drive 5014m (about 5 minutes, taxi about 24 yuan) Yantai Mountain → Fujian Normal University (Cangshan Campus): 🚲 Cycling 1432m (about 5 minutes, or about 17 minutes on foot) Fujian Normal University (Cangshan Campus) → Fuyuan Road: 🚲 Cycling 975m (about 3 minutes, or about 12 minutes on foot) Fuyuan Road → Gaogai Mountain Park: 🚗 4615m by car (about 6 minutes, taxi about 23 yuan) Gaogai Mountain Park → Luozhou Ancient Town: 🚗 6638m by car (about 9 minutes, taxi about 28 RMB)

📊 Summary of daily traffic 🚲 Riding distance: 3899m 🚗 Vehicle trips: 4 trips, total 19,370 meters 🔪 Total travel distance: 23,269 m 💰 Estimated total cost: 94 yuan (taxi and metro)

🕐 Expected timeline

🕐 Expected timeline

09:00-10:00 Straits Cultural and Arts Center (Starting point | 60min stop | Open 09:00-18:00) 10:06-11:06 Flower Sea Park (🚗 6min | Stop 60min | Opening 09:00-18:00) 11:18-12:18 Taijiang Yacht Marina (🚗 12min | 60min stop | Open 09:00-18:00) 12:28-13:28 Yantai Mountain (🚗 10min | 60min stop | Open 09:00-18:00) 13:34-14:34 Fujian Normal University (Cangshan Campus) (🚲 6min | 60min stop | Opening 09:00-18:00) 14:38-15:38 Fuyuan Road (🚲 4min | 60min stop | Open 09:00-18:00) 15:47-16:47 Gaogai Mountain Park (🚗 9min | Stop 60min | Opening 09:00-18:00) 17:00-18:00 Luozhou Ancient Town (🚗 13min | Stop 60min | Opening 09:00-18:00) 🏁 18:00 End of itinerary

Example

Straits Cultural and Arts Center

Straits Cultural and Arts Center: Five-petal jasmine flowers painted by Finnish masters, pure white curved surfaces blooming by the Min River. Even without going to see the show, just walking around the building and the moment light and shadow flow on the curved surface is enough to let you experience Fuzhou's cutting-edge urban aesthetics.

Flower Sea Park

Flower Sea Park: Seasonal flower fields embrace the river breeze, while autumn sunflowers and gesang flowers intertwine into oil painting-like color blocks. Walking along the boardwalk to the riverbank and watching the ship slowly pass by, it's a gentle corner that city dwellers have stolen from their fleeting lives.

Taijiang Yacht Marina

Taijiang Yacht Marina: Climb onto the pier boardwalk and enjoy a panoramic view of the skyline on both sides of the Min River. If the afternoon sun is just right, you can see the red brick buildings of Zhongzhou Island conversing with distant modern high-rises, and occasionally egrets skimming over the water, kicking off the riverside walk.

Yantai Mountain

Yantai Mountain: The weathered stone walls of the former consulate, the stained glass of the stone church, and the old villas lined with bougainvillea in winding alleys—every step echoes the tide of international merchant ships from a century ago. From the summit, one overlooks the Min River, where old and new Fuzhou meet.

Fujian Normal University (Cangshan Campus)

Fujian Normal University (Cangshan Campus): Hidden beneath the shadows of the century-old phoenix trees lies the red brick building of the former South China Women's College site. In the afternoon, students passed by in small groups, and the benches under the shade and the archway of the old auditorium slowed down time.

The garden road was restored

Fuyuan Road: Just a few hundred meters of lush greenery, where the Republic of China-era villa and the former consulate site stand quietly. There is no bustle of shops, only light and shadow roaming the old walls, like an unread poem, perfect for a leisurely stroll.

Gaogai Mountain Park

Gaogai Mountain Park: Climbing the steps takes less than twenty minutes, and the flying eaves of Miaofeng Temple at the summit peek out. Standing on the viewing platform, you can gaze at the Wulong River like a ribbon; the city unfolds beneath your feet, and the cool evening breeze will blow away the fatigue of the whole day.

Luozhou Ancient Town

Luozhou Ancient Town: The stone slabs of Longjiang Bridge have been polished smooth by time, and the flying eaves of the Chen family ancestral hall tell the glory of a family of imperial exams. At dusk, elderly people chat idly under the banyan tree, smoke drifts from the fishing port, and the ancient town's breath is real and slow.

Example

Day 1 map Day 2 map



Experimental Setup

04 EVALUATION

Test set

- **9 natural-language queries × 4 cities**

  Fuzhou · Guangzhou · Shenzhen · Shanghai
- 3-day plans
- Query scenarios:
  - Distance constraint (A1–A2)
  - Time minimization (A3)
  - Time-window scheduling (B1–B2)
  - Contextual awareness (C1–C2)
  - Spatial coherence (D1–D2)

Baselines & ours

- Bare LLM — no retrieval, optimization, or RAG.
- Baseline agent — has search but no spatial optimization.
- **ROAM (ours) — adds retrieval + clustering + SA-TSP + time-window correction + RAG.**
- This isolates the value of the optimizer & replanning loop.

Evaluation Metrics 6 Used

04 EVALUATION

POI ↑

Total POIs

Σ |itinerary_d|

More stops = richer route

SM ↑

Semantic Match

mean cos(e_poi, e_query)

Alignment with intent

TT ↓

Transit Ratio

commute / (avail·days) ×100%

Lower = tighter schedule

CE ↑

City Exploration

Hull(trip) / Hull(city) ×100%

Broader spatial coverage

DB

Day Balance

σ / μ of daily POIs

Lower = even daily load

RC

Route Compactness

weighted_dist / (POIs − days)

Lower = less travel cost

Comparison · Sunny&Rainy Day Scenario

04 EVALUATION

Table 1: Rainy Day Scenario Comparison (Flash & Pro)

		Fuzhou			Guangzhou			Shenzhen			Shanghai		
Model	Metric	ROAM	Bare	Baseline	ROAM	Bare	Baseline	ROAM	Bare	Baseline	ROAM	Bare	Baseline
Flash	POI↑	33	18	16	33	18	19	24	15	15	15	20	18
	SM↑	0.3	0.3	0.2	0.3	0.3	0.3	0.3	0.3	0.3	0.3	0.3	0.3
	TT↓	5.6	3.4	0.0	7.0	4.4	0.0	6.4	5.6	0.0	2.4	3.8	0.0
	CE↑	0.4	0.4	0.0	1.0	0.8	0.6	5.3	1.7	0.6	0.3	0.7	0.4
	DB	0.0	0.2	0.1	0.0	0.2	0.1	0.0	0.0	0.0	0.6	0.1	0.0
	RC	454.4	421.9	1360.4	555.5	536.0	1905.1	620.3	853.2	1586.5	316.6	428.3	1371.4
Pro	POI↑	17	19	17	16	17	17	33	16	16	15	18	18
	SM↑	0.3	0.2	0.2	0.3	0.3	0.3	0.3	0.3	0.3	0.3	0.3	0.3
	TT↓	3.2	4.4	0.0	4.7	4.9	0.0	9.6	5.0	0.0	7.3	3.6	0.0
	CE↑	0.1	0.4	0.1	0.6	3.7	0.5	32.2	9.7	0.5	3.3	0.6	0.7
	DB	0.1	0.1	0.2	0.1	0.1	0.2	0.0	0.1	0.1	0.0	0.3	0.2
	RC	402.4	507.0	686.3	643.6	635.1	1776.2	1072.5	720.3	1120.4	1009.9	460.5	1915.7

Table 2: Full Detailed Results (Flash)

Query	Metric	Fuzhou			Guangzhou			Shenzhen			Shanghai		
		ROAM	Bare	Baseline	ROAM	Bare	Baseline	ROAM	Bare	Baseline	ROAM	Bare	Baseline
Sunny	POI↑	28	21	16	21	20	19	28	17	18	33	19	15
	SM↑	0.3	0.3	0.3	0.3	0.4	0.4	0.4	0.3	0.4	0.3	0.4	0.4
	TT↓	7.6	3.9	0.0	**6.8**	8.2	0.0	12.2	6.9	0.0	6.6	3.2	0.0
	CE↑	0.5	0.3	1.0	0.7	4.8	0.6	27.7	37.5	1.2	1.1	0.4	0.2
	DB	0.1	0.1	0.1	0.1	0.1	0.2	0.2	0.1	0.0	0.0	0.1	0.0
	RC	554.8	411.8	3856.9	626.3	869.6	1433.9	1043.2	872.4	1716.1	483.3	379.0	1396.4
Rainy	POI↑	33	18	16	33	18	19	24	15	15	15	20	18
	SM↑	0.3	0.3	0.2	0.3	0.3	0.3	0.3	0.3	0.3	0.3	0.3	0.3
	TT↓	5.6	3.4	0.0	7.0	4.4	0.0	6.4	5.6	0.0	2.4	3.8	0.0
	CE↑	**0.4**	**0.4**	0.0	**1.0**	0.8	0.6	**5.3**	1.7	0.6	0.3	0.7	0.4
	DB	0.0	0.2	0.1	0.0	0.2	0.1	0.0	0.0	0.0	0.6	0.1	0.0
	RC	454.4	421.9	1360.4	555.5	536.0	1905.1	620.3	853.2	1586.5	316.6	428.3	1371.4
Sightseeing	**POI↑**	**36**	16	18	**35**	15	19	**34**	15	16	**33**	16	16
	SM↑	0.4	0.4	0.4	0.4	0.4	0.4	0.4	0.4	0.4	0.5	0.5	0.5
	TT↓	8.7	6.7	0.0	12.0	5.2	0.0	8.6	4.5	0.0	13.8	3.8	0.0
	CE↑	1.6	0.7	0.0	6.1	2.3	1.1	9.8	4.4	0.5	29.7	0.4	0.1
	DB	0.0	0.1	0.2	0.0	0.0	0.2	0.1	0.0	0.1	0.2	0.1	0.1
	RC	710.8	933.2	526.3	889.1	786.9	1932.6	720.2	675.0	1765.6	1653.1	550.6	1528.8
City Exploration	POI↑	24	20	19	23	18	15	25	15	18	21	19	16
	SM↑	0.3	0.3	0.3	0.3	0.3	0.3	0.3	0.3	0.3	0.2	0.3	0.3
	TT↓	4.8	3.6	0.0	4.9	4.1	0.0	7.6	4.6	0.0	5.7	3.4	0.0
	CE↑	**0.3**	**0.3**	0.1	**0.7**	0.6	0.3	5.4	16.4	2.9	**0.7**	0.6	0.2
	DB	0.7	0.1	0.2	0.1	0.0	0.0	0.1	0.2	0.2	0.0	0.1	0.1
	RC	404.0	392.6	621.1	426.9	505.9	1436.6	583.5	704.6	2095.4	514.6	403.5	1415.6

Failure Cases & Discussion

05 · DISCUSSION

When it fails

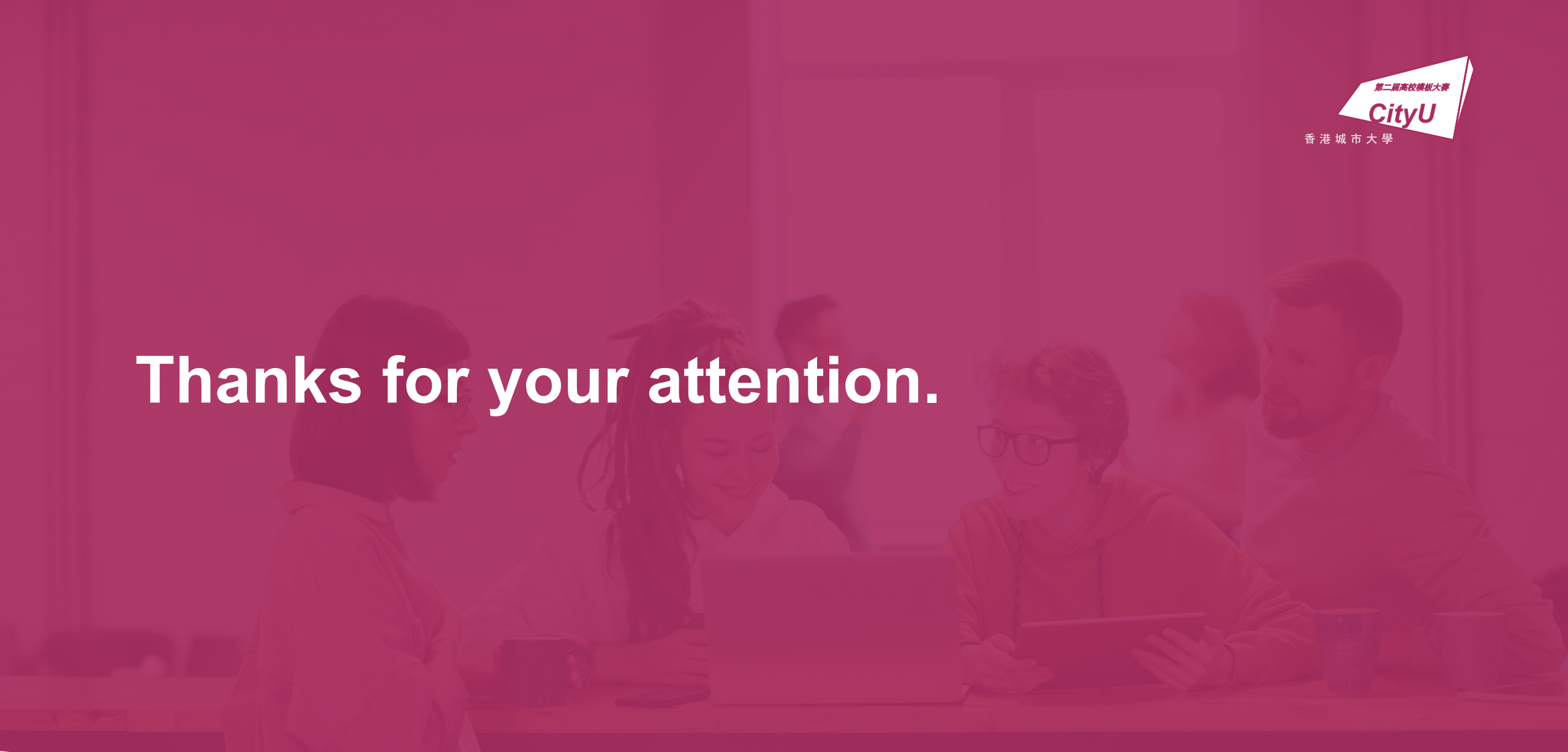
- Ambiguous requests that need clarification the LLM must ask back.
- No matching POI (closed / unavailable) for the requested category.
- Conflicting constraints — too many stops, impossible time windows.
- Cost & latency: each replan uses several LLM turns + tool calls; tokens grow with plan size.

Limitations & future work

- Evaluation is on offline queries; live interactive latency not yet measured.
- Real-time adaptation — live traffic / queue times.
- Personalization loop from user feedback.
- Multi-city and multi-traveler support.

第二屆高校模板大賽
CityU
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Thanks for your attention.



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